Challenge Problem Set 2

Math 292 Spring 2026
Rutgers University

Exercise 1:

1. We can differentiate our solution x(t) to get

$$\mathbf{x}''(t) = \sum_{j=0}^{4} x_j''(t)$$

2. Thus we have for x_0 that:

$$x_0''(t) = -Kx_0(t)$$

3. And we have for $j \in \{1, 2, 3, 4\}$ that

$$x_j''(t) = -Kx_j(t) + f_j(t)$$

4. Combining the two previous equations we have

$$\begin{aligned}\mathbf{x}''(t) &= -Kx_0(t) + \sum_{j=1}^{4}(-Kx_j(t) + f_j(t)) \\ &= -K\sum_{i=0}^{4}(x_i(t)) + \sum_{j=1}^{4}(f_j(t)) \\ &= -K\mathbf{x}(t) + \mathbf{f}(t)\end{aligned}$$

5. This is exactly the original differential equation.
6. The initial conditions also check out

$$\begin{aligned}\mathbf{x}(0) &= \sum_{j=0}^{4}(x_j(t)) \\ &= (1,2) + (0,0) + (0,0) + (0,0) + (0,0) \\ &= (1,2)\end{aligned}$$

$$\begin{aligned}\mathbf{x}'(0) &= \sum_{j=0}^{4}(x_j'(t)) \\ &= (1,1) + (0,0) + (0,0) + (0,0) + (0,0) \\ &= (1,1)\end{aligned}$$

Exercise 2:

1. We begin by diagonalizing K in order the uncouple the system. To do this, we first need to solve for the eigenvalues using the characteristic polynomial of K.

$$\begin{aligned} det\begin{pmatrix} 3-\lambda & 2 \\ 2 & 6-\lambda \end{pmatrix} &= (3-\lambda)(6-\lambda)-4 \\ &= \lambda^2 - 9\lambda + 14 \\ &= (\lambda-7)(\lambda-2) \\ &= 0 \end{aligned}$$

2. We now have that the eigenvalues are

$$\lambda_1 = 7,\ \lambda_2 = 2$$

3. With this, we can solve for the eigenvectors of K

- $\lambda_1 = 7$:

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \mathbf{v}_1 = \mathbf{0}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- $\lambda_1 = 2$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \mathbf{v}_2 = \mathbf{0}$$

$$\mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

4. Thus, using the diagonalization of K, we have

$$\begin{aligned} \mathbf{x}''(t) &= -VSV^{-1}\mathbf{x}(t) \\ &= -\frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 7 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}\mathbf{x}(t) \end{aligned}$$

5. We then define a new variable y

$$\mathbf{y} = V^{-1}\mathbf{x}(t) = \frac{1}{\sqrt{5}}\begin{bmatrix} 1x_1 + 2x_2 \\ -2x_1 + 1x_2 \end{bmatrix}$$

6. Rearranging the equation we have

$$\begin{aligned} \mathbf{x}''(t) &= -VS\mathbf{y}(\mathbf{t}) \\ V^{-1}\mathbf{x}''(t) &= -S\mathbf{y}(\mathbf{t}) \\ \mathbf{y}'' &= -S\mathbf{y} \end{aligned}$$

7. We have the following system

$$y_1'' = -7y_1$$
$$y_2'' = -2y_2$$

8. This is the same as the solutions of the previous workshop, thus we have

$$y_1(t) = c_1\cos(\sqrt{7}t) + c_2\sin(\sqrt{7}t)$$
$$y_2(t) = c_3\cos(\sqrt{2}t) + c_4\sin(\sqrt{2}t)$$
$$y_1'(t) = \sqrt{7}c_2\cos(\sqrt{7}t) - \sqrt{7}c_1\sin(\sqrt{7}t)$$
$$y_2'(t) = \sqrt{2}c_4\cos(\sqrt{2}t) - \sqrt{2}c_3\sin(\sqrt{2}t)$$
$$\text{with } c_1, c_2, c_3, c_4 \in \mathbb{R}$$

9. At $t = 0$ this would turn into

$$y_1(0) = c_1$$
$$y_2(0) = c_3$$
$$y_1'(0) = \sqrt{7}c_2$$
$$y_2'(0) = \sqrt{2}c_4$$

9. We use the change of basis equation and initial conditions to solve for our constants

$$\begin{aligned}\mathbf{y}(0) &= V^{-1}\mathbf{x}(0)\\ &= \frac{1}{\sqrt{5}}\begin{pmatrix}1 & 2\\ -2 & 1\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix}\\ &= \frac{1}{\sqrt{5}}\begin{pmatrix}5\\ 0\end{pmatrix}\end{aligned}$$

10. We have our first two constants

$$c_1 = \sqrt{5}$$
$$c_3 = 0$$

11. Now, we repeat with y'

$$\begin{aligned}\mathbf{y}'(0) &= V^{-1}\mathbf{x}'(0)\\ &= \frac{1}{\sqrt{5}}\begin{pmatrix}1 & 2\\ -2 & 1\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix}\\ &= \frac{1}{\sqrt{5}}\begin{pmatrix}3\\ -1\end{pmatrix}\end{aligned}$$

12. Solving the system we have

$$c_2 = \frac{3}{\sqrt{35}}$$
$$c_4 = -\frac{1}{\sqrt{10}}$$

13. Finally, we have the final solution:

$$\begin{aligned}\mathbf{x}_0(t) &= V\mathbf{y}(t)\\ &= y_1(t)\mathbf{v_1} + y_2(t)\mathbf{v_2}\\ &= (\cos(\sqrt{7}t) + \frac{3}{5\sqrt{7}}\sin(\sqrt{7}t))\begin{bmatrix}1\\2\end{bmatrix} - (\frac{1}{5\sqrt{2}}\sin(\sqrt{2}t))\begin{bmatrix}-2\\1\end{bmatrix}\end{aligned}$$

14. Separating $\mathbf{x}_0$ into it's components we have

$$\begin{aligned}x_1(t) &= \cos(\sqrt{7}t) + \frac{3}{5\sqrt{7}}sin(\sqrt{7}t) + \frac{2}{5\sqrt{2}}\sin(\sqrt{2}t)\\ x_2(t) &= 2\cos(\sqrt{7}t) + \frac{6}{5\sqrt{7}}sin(\sqrt{7}t) - \frac{1}{5\sqrt{2}}\sin(\sqrt{2}t)\end{aligned}$$

Exercise 3:

Now we have that $\mathbf{x}_1''(t) = -K\mathbf{x}_1(t) + \mathbf{f}_1(t)$, but we can again use the diagonalization of K from the previous part to solve for the solution,

$$\begin{aligned}\mathbf{x}_1''(t) &= -K\mathbf{x}_1(t) + \mathbf{f}_1(t)\\ &= -VSV^{-1}\mathbf{x}_1(t) = \mathbf{f}_1(t)\end{aligned}$$

We let $\mathbf{y}(t) = V^{-1}\mathbf{x}(t)$ and multiply both sides by V^{-1}

$$\mathbf{y}''(t) + S\mathbf{y}(t) = V^{-1}\mathbf{f}_1(t)$$

We are left with the following system

$$\begin{aligned}y_1''(t) + 7y_1 &= \frac{7}{\sqrt{5}}\cos(\omega_1 t)\\ y_2''(t) + 2y_2 &= \frac{1}{\sqrt{5}}\cos(\omega_1 t)\end{aligned}$$

Similar to the last part, we guess the solution to y

$$\begin{aligned}y_1(t) &= A_1\cos(\sqrt{7}t) + B_1\sin(\sqrt{7}t) + C_1 cos(\omega_1 t)\\ y_2(t) &= A_2\cos(\sqrt{2}t) + B_2\sin(\sqrt{2}t) + C_2 cos(\omega_1 t)\end{aligned}$$

Then, taking the first and second derivative we have

$$\begin{aligned}y_1'(t) &= -\sqrt{7}A_1\sin(\sqrt{7}t) + \sqrt{7}B_1\cos(\sqrt{7}t) - \omega_1 C_1 sin(\omega_1 t)\\ y_2'(t) &= -\sqrt{2}A_2\sin(\sqrt{2}t) + \sqrt{2}B_2\cos(\sqrt{2}t) - \omega_1 C_2 sin(\omega_1 t)\end{aligned}$$

$$\begin{aligned}y_1''(t) &= -7A_1\cos(\sqrt{7}t) - 7B_1\sin(\sqrt{7}t) - \omega_1^2 C_1 cos(\omega_1 t)\\ y_2''(t) &= -2A_2\cos(\sqrt{2}t) - 2B_2\sin(\sqrt{2}t) - \omega_1^2 C_2 cos(\omega_1 t)\end{aligned}$$

We solve for C_1 and C_2 by plugging into the first system we solved for

- For C_1:

$$\begin{aligned}-\omega_1^2 C_1\cos(\omega_1 t) + 7C_1 cos(\omega_1 t) &= \frac{7}{\sqrt{5}}cos(\omega_1 t)\\ C_1(7-\omega_1^2) &= \frac{7}{\sqrt{5}}\\ C_1 &= \frac{7}{\sqrt{5}(7-\omega_1^2)}\end{aligned}$$

- For C_2:

$$\begin{aligned} -\omega_1^2 C_2 \cos(\omega_1 t) + 2C_2 cos(\omega_1 t) &= \frac{1}{\sqrt{5}} cos(\omega_1 t) \\ C_2(2 - \omega_1^2) &= \frac{1}{\sqrt{5}} \\ C_2 &= \frac{1}{\sqrt{5}(2 - \omega_1^2)} \end{aligned}$$

Next, we solve for A_1 and A_2 with the initial condition $\mathbf{x}(0) = 0 \implies \mathbf{y}(0) = 0$

- For A_1:

$$\begin{aligned} y_1(0) &= A_1 \cos(0) + B_1 \sin(0) + C_1 cos(0) \\ A_1 &= -C_1 \\ &= -\frac{7}{\sqrt{5}(7 - \omega_1^2)} \end{aligned}$$

- For A_2:

$$\begin{aligned} y_2(0) &= A_2 \cos(0) + B_2 \sin(0) + C_2 cos(0) \\ A_2 &= -C_2 \\ &= -\frac{1}{\sqrt{5}(2 - \omega_1^2)} \end{aligned}$$

We repeat this again for B_1 and B_2 using the initial condition $\mathbf{x}'(0) = 0 \implies \mathbf{y}'(0) = 0$

- For B_1:

$$\begin{aligned} y_1'(0) &= -\sqrt{7} A_1 \sin(0) + \sqrt{7} B_1 \cos(0) - \omega_1 C_1 sin(0) \\ B_1 &= 0 \end{aligned}$$

- For B_2:

$$\begin{aligned} y_2'(0) &= -\sqrt{2} A_2 \sin(0) + \sqrt{2} B_2 \cos(0) - \omega_1 C_2 sin(0) \\ B_2 &= 0 \end{aligned}$$

Notice that B_1 and B_2 will always be 0 for $x_i \mid i = 1, 2, 3, 4$ since only the middle term will not go to zero when t=0.

Thus, we have our solution for $\mathbf{y}(t)$, where $\omega_i(t) \neq \sqrt{7}, \sqrt{2} \mid i = 1, 2, 3, 4$

$$\mathbf{y}(t) = \begin{bmatrix} \frac{7}{\sqrt{5}(7 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7} t)) \\ \frac{1}{\sqrt{5}(2 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{2} t)) \end{bmatrix}$$

Converting back to our original basis with $\mathbf{x}(t) = V\mathbf{y}(t)$, we get

$$\mathbf{x_1}(t) = \begin{bmatrix} \frac{7}{5(7 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7} t)) - \frac{2}{5(2 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{2} t)) \\ \frac{14}{5(7 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7} t)) + \frac{1}{5(2 - \omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{2} t)) \end{bmatrix}$$

However, we must also consider the case where $\omega_1 = \pm\sqrt{7}$ or $\omega_1 = \pm\sqrt{2}$ or when the system is in "resonance". Choosing the positive values, we have

- $\omega_1 = \sqrt{7}$:

  We take the limit of $y_1(t)$ as the ω_1 approaches the natural frequency, via L'Hopital's Rule

$$\lim_{\omega_1 \to \sqrt{7}} \frac{7}{\sqrt{5}(7-\omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7}t)) \Rightarrow \frac{-7t\sin(\omega_1 t)}{-2\sqrt{5}\omega_1}$$
$$\Rightarrow \frac{\sqrt{7}}{2\sqrt{5}} t\sin(\sqrt{7}t)$$

  Then, we do the same for $y_2(t)$ by just substituting in the limit

$$\lim_{\omega_2 \to \sqrt{7}} \frac{1}{\sqrt{5}(2-\omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{2}t)) \Rightarrow \frac{1}{5\sqrt{5}}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t))$$

  Thus, for the final solution, we have that

$$\mathbf{x_1}(t) = \begin{bmatrix} \frac{\sqrt{7}}{10} t\sin(\sqrt{7}t) - \frac{2}{25}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t)) \\ \frac{\sqrt{7}}{5} t\sin(\sqrt{7}t) + \frac{1}{25}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t)) \end{bmatrix}$$

- $\omega_1 = \sqrt{2}$

  We take the limit of $y_1(t)$ as the ω_1 approaches the natural frequency, substituting for $\sqrt{2}$

$$\lim_{\omega_1 \to \sqrt{2}} \frac{7}{\sqrt{5}(7-\omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7}t)) \Rightarrow \frac{7}{5\sqrt{5}}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t))$$

  Then, we do the same for $y_2(t)$ via L'Hopital's Rule

$$\lim_{\omega_2 \to \sqrt{2}} \frac{\cos(\omega_1 t) - \cos(\sqrt{2}t)}{\sqrt{5}(2-\omega_1^2)} \Rightarrow \frac{-t\sin(\omega_1 t)}{-2\sqrt{5}\omega_1}$$
$$\Rightarrow \frac{1}{2\sqrt{10}} t\sin(\sqrt{2}t)$$

  Thus, for the final solution, we have that

$$\mathbf{x_1}(t) = \begin{bmatrix} \frac{7}{25}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t)) - \frac{1}{5\sqrt{2}} t\sin(\sqrt{2}t) \\ \frac{14}{25}(\cos(\sqrt{2}t) - \cos(\sqrt{7}t)) + \frac{1}{10\sqrt{2}} t\sin(\sqrt{2}t) \end{bmatrix}$$

Exercise 4:

We follow a similar process, to the last part, the only different part is f(t). Thus, we start with the change of basis $\mathbf{y}(t) = V^{-1}\mathbf{x}(t)$

$$\mathbf{y}''(t) + S\mathbf{y}(t) = V^{-1}\mathbf{f}_2(t)$$

This gives us the following system

$$y_1''(t) + 7y_1 = -\frac{1}{\sqrt{5}}\sin(\omega_2 t)$$
$$y_2''(t) + 2y_2 = -\frac{3}{\sqrt{5}}\sin(\omega_2 t)$$

We guess the solution of y to be

$$y_1(t) = A_1\cos(\sqrt{7}t) + B_1\sin(\sqrt{7}t) + C_1\sin(\omega_2 t)$$
$$y_2(t) = A_2\cos(\sqrt{2}t) + B_2\sin(\sqrt{2}t) + C_2\sin(\omega_2 t)$$

Then, taking the first and second derivative we have

$$y_1'(t) = -\sqrt{7}A_1\sin(\sqrt{7}t) + \sqrt{7}B_1\cos(\sqrt{7}t) + \omega_2 C_1\cos(\omega_2 t)$$
$$y_2'(t) = -\sqrt{2}A_2\sin(\sqrt{2}t) + \sqrt{2}B_2\cos(\sqrt{2}t) + \omega_2 C_2\cos(\omega_2 t)$$

$$y_1''(t) = -7A_1\cos(\sqrt{7}t) - 7B_1\sin(\sqrt{7}t) - \omega_2^2 C_1\sin(\omega_2 t)$$
$$y_2''(t) = -2A_2\cos(\sqrt{2}t) - 2B_2\sin(\sqrt{2}t) - \omega_2^2 C_2\sin(\omega_2 t)$$

We solve for C_1 and C_2 by plugging into the first system we solved for

- For C_1:

$$-\omega_2^2 C_1\sin(\omega_2 t) + 7C_1\sin(\omega_2 t) = -\frac{1}{\sqrt{5}}sin(\omega_2 t)$$
$$C_1(7-\omega_2^2) = -\frac{1}{\sqrt{5}}$$
$$C_1 = \frac{-1}{\sqrt{5}(7-\omega_2^2)}$$

- For C_2:

$$-\omega_2^2 C_2\sin(\omega_2 t) + 2C_2\sin(\omega_2 t) = -\frac{3}{\sqrt{5}}\sin(\omega_2 t)$$
$$C_2(2-\omega_2^2) = -\frac{3}{\sqrt{5}}$$
$$C_2 = \frac{-3}{\sqrt{5}(2-\omega_2^2)}$$

Next, we solve for A_1 and A_2 with the initial condition $\mathbf{x}(0) = 0 \implies \mathbf{y}(0) = 0$

- For A_1:

$$y_1(0) = A_1\cos(\sqrt{7}t) + B_1\sin(0) + C_1\sin(0)$$
$$A_1 = 0$$

- For A_2:

$$y_2(0) = A_2\cos(0) + B_2\sin(0) + C_2\sin(0)$$
$$A_2 = 0$$

From this and the previous exercise, we can see that when $\mathbf{f}(t)$ is a multiple of $\cos(t)$, then the solution of y reduces to

$$\mathbf{y} = \begin{bmatrix} A_1\cos(\sqrt{7}t) + C_1 cos(\omega_i t) \\ A_2\cos(\sqrt{2}t) + C_2 cos(\omega_i t) \end{bmatrix} \mid i = 1, 4$$

and when $\mathbf{f}(t)$ is a multiple of $\sin(t)$, then the solution of y reduces to

$$\mathbf{y} = \begin{bmatrix} B_1 \sin(\sqrt{7}t) + C_1 \sin(\omega_i t) \\ B_2 \sin(\sqrt{2}t) + C_2 \sin(\omega_i t) \end{bmatrix} \mid i = 2, 3$$

We can reuse this fact in the next two exercises.
Now, we solve for B_1 and B_2 using the initial condition $\mathbf{x}'(0) = 0 \implies \mathbf{y}'(0) = 0$

- For B_1:

$$\begin{aligned} y_1'(0) &= -\sqrt{7}A_1 \sin(0) + \sqrt{7}B_1 \cos(0) + \omega_2 C_1 \cos(0) \\ B_1 &= -\frac{\omega_2 C_1}{\sqrt{7}} \\ &= \frac{\omega_2}{\sqrt{35}(7 - \omega_2^2)} \end{aligned}$$

- For B_2:

$$\begin{aligned} y_2'(0) &= -\sqrt{2}A_2 \sin(0) + \sqrt{2}B_2 \cos(0) + \omega_2 C_2 \cos(0) \\ B_2 &= -\frac{\omega_2 C_2}{\sqrt{2}} \\ &= \frac{3\omega_2}{\sqrt{10}(2 - \omega_2^2)} \end{aligned}$$

Thus, we have our solution for $\mathbf{y}(t)$,

$$\mathbf{y} = \begin{bmatrix} \frac{1}{\sqrt{5}(7-\omega_2^2)}(\frac{\omega_2}{\sqrt{7}} \sin(\sqrt{7}t) - \sin(\omega_2 t)) \\ \frac{3}{\sqrt{5}(2-\omega_2^2)}(\frac{\omega_2}{\sqrt{2}} \sin(\sqrt{2}t) - \sin(\omega_2 t)) \end{bmatrix}$$

Converting back to our original basis with $\mathbf{x}(t) = V\mathbf{y}(t)$, we get

$$\mathbf{x_2}(t) = \begin{bmatrix} \frac{1}{5(7-\omega_2^2)}(\frac{\omega_2}{\sqrt{7}} \sin(\sqrt{7}t) - \sin(\omega_2 t)) - \frac{6}{5(2-\omega_2^2)}(\frac{\omega_2}{\sqrt{2}} \sin(\sqrt{2}t) - \sin(\omega_2 t)) \\ \frac{2}{5(7-\omega_2^2)}(\frac{\omega_2}{\sqrt{7}} \sin(\sqrt{7}t) - \sin(\omega_2 t)) + \frac{3}{5(2-\omega_2^2)}(\frac{\omega_2}{\sqrt{2}} \sin(\sqrt{2}t) - \sin(\omega_2 t)) \end{bmatrix}$$

Exercise 5:

We follow a similar process, to the last part, the only different part is f(t). Thus, we start with the change of basis $\mathbf{y}(t) = V^{-1}\mathbf{x}(t)$

$$\mathbf{y}''(t) + S\mathbf{y}(t) = V^{-1}\mathbf{f}_3(t)$$

This gives us the following system

$$\begin{aligned} y_1''(t) + 7y_1 &= \frac{1}{\sqrt{5}} \sin(\omega_3 t) \\ y_2''(t) + 2y_2 &= -\frac{7}{\sqrt{5}} \sin(\omega_3 t) \end{aligned}$$

We guess the solution of y to be

$$\begin{aligned} y_1(t) &= B_1 \sin(\sqrt{7}t) + C_1 \sin(\omega_3 t) \\ y_2(t) &= B_2 \sin(\sqrt{2}t) + C_2 \sin(\omega_3 t) \end{aligned}$$

Then, taking the first and second derivative we have

$$y_1'(t) = \sqrt{7}B_1\cos(\sqrt{7}t) + \omega_3 C_1\cos(\omega_3 t)$$
$$y_2'(t) = \sqrt{2}B_2\cos(\sqrt{2}t) + \omega_3 C_2\cos(\omega_3 t)$$

$$y_1''(t) = -7B_1\sin(\sqrt{7}t) - \omega_3^2 C_1\sin(\omega_3 t)$$
$$y_2''(t) = -2B_2\sin(\sqrt{2}t) - \omega_3^2 C_2\sin(\omega_3 t)$$

We solve for C_1 and C_2 by plugging into the first system we solved for

- For C_1:

$$\begin{aligned} -\omega_3^2 C_1\sin(\omega_3 t) + 7C_1\sin(\omega_3 t) &= \frac{1}{\sqrt{5}}\sin(\omega_3 t) \\ C_1(7-\omega_3^2) &= \frac{1}{\sqrt{5}} \\ C_1 &= \frac{1}{\sqrt{5}(7-\omega_3^2)} \end{aligned}$$

- For C_2:

$$\begin{aligned} -\omega_3^2 C_2\sin(\omega_3 t) + 2C_2\sin(\omega_3 t) &= -\frac{7}{\sqrt{5}}\sin(\omega_3 t) \\ C_2(2-\omega_3^2) &= -\frac{7}{\sqrt{5}} \\ C_2 &= \frac{-7}{\sqrt{5}(2-\omega_3^2)} \end{aligned}$$

Now, we solve for B_1 and B_2 using the initial condition $\mathbf{x}'(0) = 0 \implies \mathbf{y}'(0) = 0$

- For B_1:

$$\begin{aligned} B_1 &= -\frac{\omega_3 C_1}{\sqrt{7}} \\ &= \frac{-\omega_3}{\sqrt{35}(7-\omega_3^2)} \end{aligned}$$

- For B_2:

$$\begin{aligned} B_2 &= -\frac{\omega_3 C_2}{\sqrt{2}} \\ &= \frac{7\omega_3}{\sqrt{10}(2-\omega_3^2)} \end{aligned}$$

Thus, we have our solution for $\mathbf{y}(t)$,

$$\mathbf{y} = \begin{bmatrix} \frac{1}{\sqrt{5}(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) \\ \frac{-7}{\sqrt{5}(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \end{bmatrix}$$

Converting back to our original basis with $\mathbf{x}(t) = V\mathbf{y}(t)$, we get

$$\mathbf{x_3}(t) = \begin{bmatrix} \frac{1}{5(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) + \frac{14}{5(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \\ \frac{2}{5(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) + \frac{-7}{5(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \end{bmatrix}$$

Exercise 6:

We follow a similar process, to the last part, the only different part is f(t). Thus, we start with the change of basis $\mathbf{y}(t) = V^{-1}\mathbf{x}(t)$

$$\mathbf{y}''(t) + S\mathbf{y}(t) = V^{-1}\mathbf{f}_4(t)$$

This gives us the following system

$$y_1''(t) + 7y_1 = \frac{1}{\sqrt{5}}\cos(\omega_4 t)$$
$$y_2''(t) + 2y_2 = -\frac{2}{\sqrt{5}}\cos(\omega_4 t)$$

We guess the solution of y to be

$$y_1(t) = A_1\cos(\sqrt{7}t) + C_1 cos(\omega_4 t)$$
$$y_2(t) = A_2\cos(\sqrt{2}t) + C_2 cos(\omega_4 t)$$

Then, taking the first and second derivative we have

$$y_1'(t) = -\sqrt{7}A_1\sin(\sqrt{7}t) - \omega_4 C_1 sin(\omega_4 t)$$
$$y_2'(t) = -\sqrt{2}A_2\sin(\sqrt{2}t) - \omega_4 C_2 sin(\omega_4 t)$$

$$y_1''(t) = -7A_1\cos(\sqrt{7}t) - \omega_4^2 C_1 cos(\omega_4 t)$$
$$y_2''(t) = -2A_2\cos(\sqrt{2}t) - \omega_4^2 C_2 cos(\omega_4 t)$$

We solve for C_1 and C_2 by plugging into the first system we solved for

- For C_1:

$$C_1(7 - \omega_4^2) = \frac{1}{\sqrt{5}}$$
$$C_1 = \frac{1}{\sqrt{5}(7 - \omega_4^2)}$$

- For C_2:

$$C_2(2 - \omega_4^2) = -\frac{2}{\sqrt{5}}$$
$$C_2 = \frac{-2}{\sqrt{5}(2 - \omega_4^2)}$$

Next, we solve for A_1 and A_2 with the initial condition $\mathbf{x}(0) = 0 \implies \mathbf{y}(0) = 0$

- For A_1:

$$\begin{aligned} y_1(0) &= A_1\cos(0) + B_1\sin(0) + C_1 cos(0) \\ A_1 &= -C_1 \\ &= -\frac{1}{\sqrt{5}(7-\omega_4^2)} \end{aligned}$$

- For A_2:

$$\begin{aligned} y_2(0) &= A_2\cos(0) + B_2\sin(0) + C_2 cos(0) \\ A_2 &= -C_2 \\ &= -\frac{-2}{\sqrt{5}(2-\omega_4^2)} \end{aligned}$$

Thus, we have our solution for $\mathbf{y}(t)$,

$$\mathbf{y} = \begin{bmatrix} \frac{1}{\sqrt{5}(7-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{7}t)) \\ \frac{-2}{\sqrt{5}(2-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{2}t)) \end{bmatrix}$$

Converting back to our original basis with $\mathbf{x}(t) = V\mathbf{y}(t)$, we get

$$\mathbf{x_4}(t) = \begin{bmatrix} \frac{1}{5(7-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{7}t)) + \frac{4}{5(2-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{2}t)) \\ \frac{2}{5(7-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{7}t)) - \frac{2}{5(2-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{2}t)) \end{bmatrix}$$

Exercise 7:

Combining our solutions we have the final solution for (0.1) and (0.2)

$$\mathbf{x}(t) = (\frac{1}{\sqrt{5}}y_1(t))\begin{bmatrix}1\\2\end{bmatrix} + (\frac{1}{\sqrt{5}}y_2(t))\begin{bmatrix}-2\\1\end{bmatrix}$$

$$\begin{aligned}
\frac{1}{\sqrt{5}}y_1(t) &= \cos(\sqrt{7}t) + \frac{3}{5\sqrt{7}}\sin(\sqrt{7}t) \\
&\quad + \frac{7}{5(7-\omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{7}t)) \\
&\quad + \frac{1}{5(7-\omega_2^2)}(\frac{\omega_2}{\sqrt{7}}\sin(\sqrt{7}t) - \sin(\omega_2 t)) \\
&\quad + \frac{1}{5(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) \\
&\quad + \frac{1}{5(7-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{7}t)) \\
\frac{1}{\sqrt{5}}y_2(t) &= \frac{-1}{5\sqrt{2}}\sin(\sqrt{2}t) \\
&\quad + \frac{1}{5(2-\omega_1^2)}(\cos(\omega_1 t) - \cos(\sqrt{2}t)) \\
&\quad + \frac{3}{5(2-\omega_2^2)}(\frac{\omega_2}{\sqrt{2}}\sin(\sqrt{2}t) - \sin(\omega_2 t)) \\
&\quad + \frac{-7}{5(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \\
&\quad + \frac{-2}{5(2-\omega_4^2)}(\cos(\omega_4 t) - \cos(\sqrt{2}t))
\end{aligned}$$

Notice that the system will exhibit resonance when $\omega_i = \sqrt{7}$ or $\omega_i = \sqrt{2} \mid i = 1, 2, 3, 4$

Exercise 8:

The closer the ω values are to resonance, the closer the denominator will be to zero which will lead to the function to blow up.

Approximating, we have that

$$\sqrt{2} \approx 1.414$$
$$\sqrt{7} \approx 2.646$$

And the closest $\omega_i \mid i = 1, 2, 3, 4$ is $\omega_3 = 2.65$.

Therefore, the solution will still be well approximated by just keeping the $\mathbf{x_3}(t)$ term or that

$$\mathbf{x}(t) \approx \begin{bmatrix} \frac{1}{5(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) + \frac{14}{5(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \\ \frac{2}{5(7-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{7}}\sin(\sqrt{7}t)) + \frac{-7}{5(2-\omega_3^2)}(\sin(\omega_3 t) - \frac{\omega_3}{\sqrt{2}}\sin(\sqrt{2}t)) \end{bmatrix}$$

with $\omega_3 = 2.65$